

STUDENT RETENTION & WELFARE EFFICACY TRACKER

A Data-Driven Dashboard for Monitoring Attendance, Learning Outcomes, Mid-Day Meal Procurement and School Infrastructure

Competition	TransOrg AgentIQ Datathon
Track	Track 4 — Education & EdTech
Team	K Adithya Sankar
Technology	Python, Pandas, Power BI, DAX 2-page
Dashboard	interactive Power BI dashboard
Report Date	September 2026

Table of Contents

1. Executive Summary	4
2. Problem Statement	4
3. Objectives	4
4. Dataset Description	5
5. Data Preparation & Cleaning	5
5.1 Attendance Cleaning.....	. 5
5.2 Mid-Day Meal (MDM)DataPreparation.	. 6
5.3 School InfrastructureCleaning.....	. 6
5.4 Test Score Standardization.....	. 6
6. Attendance Anomaly Detection	8
7. Infrastructure Deficit Methodology	8
8. Power BI Data Model	8
9. Key DAX Measures	8
10. Dashboard Key Performance Indicators	10
11. Dashboard Page 1 — Executive Overview	11
12. Dashboard Page 2 — Risk & Welfare Deep Dive	11
13. Key Findings	13
13.1 Attendance.....	.13
13.2 Potential AttendanceAnomalies	.13
13.3 Learning Outcomes.....	.13
13.4 Mid-Day MealProcurement..	.13
13.5 Infrastructure.....	.13

14. Business / Administrative Use Cases	13
15. Limitations and Assumptions	14
15.1 Synthetic Data.....	.14
15.2 Proxy Anomalies.....	.14
15.3 Missing GenderInformation	.14
15.4 MDM Wastage.....	.14
15.5 Grading Conversion....	.14
15.6 Infrastructure History....	.14
15.7 Missing Values.....	.14
16. Future Enhancements	14
17. Conclusion	15
18. Technology Stack	15